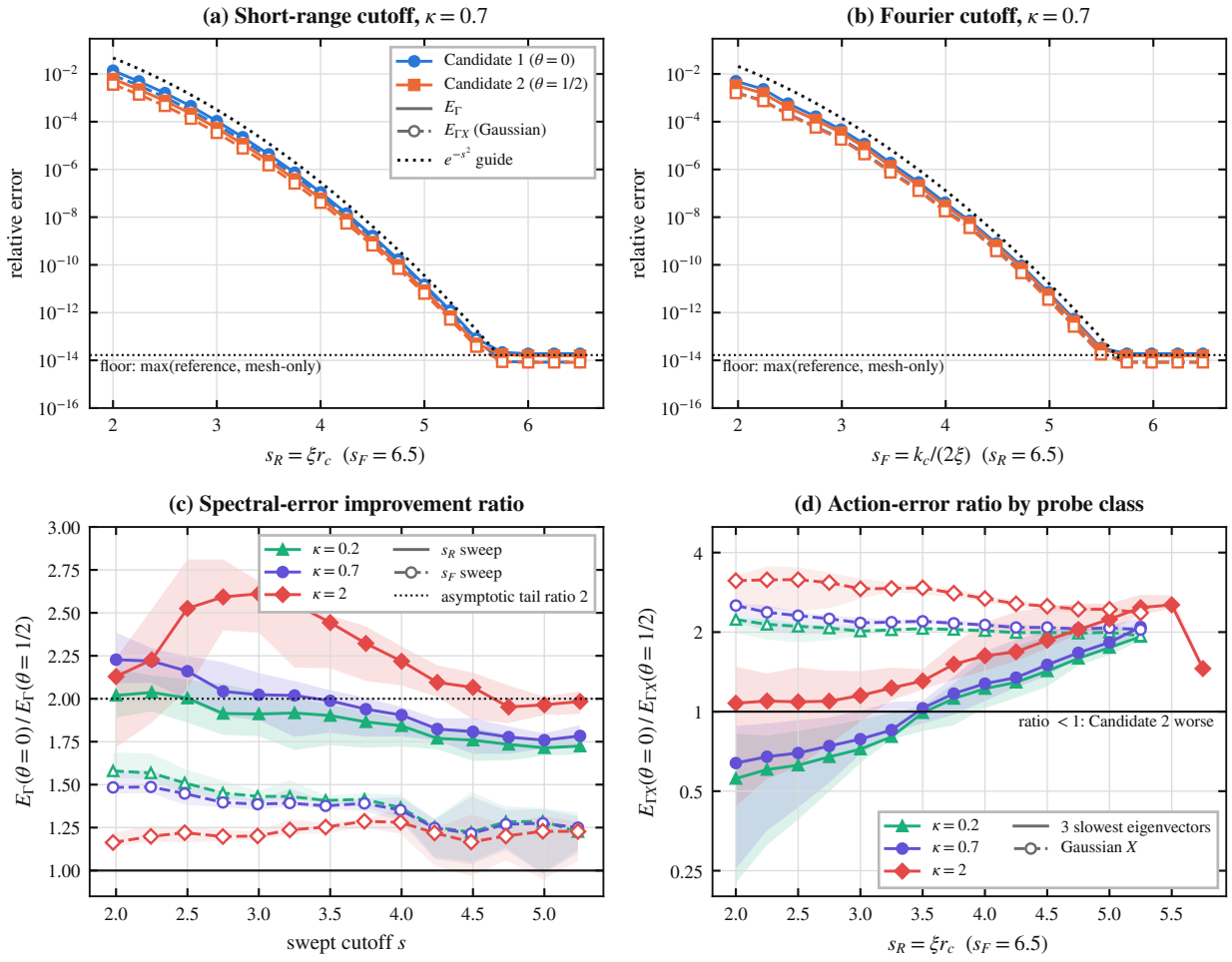
(a) Short-range cutoff, $\kappa = 0.7$
relative error
Candidate 1 ($\theta = 0$)
Candidate 2 ($\theta = 1/2$)
E_Γ
$E_{\Gamma X}$ (Gaussian)
e^{-s^2} guide
floor: max(reference, mesh-only)
$s_R = \xi r_c$ ($s_F = 6.5$)
(b) Fourier cutoff, $\kappa = 0.7$
relative error
floor: max(reference, mesh-only)
$s_F = k_c/(2\xi)$ ($s_R = 6.5$)
(c) Spectral-error improvement ratio
$E_\Gamma(\theta = 0)/E_\Gamma(\theta = 1/2)$
$\kappa = 0.2$
$\kappa = 0.7$
$\kappa = 2$
s_R sweep
s_F sweep
asymptotic tail ratio 2
swept cutoff s
(d) Action-error ratio by probe class
$E_{\Gamma X}(\theta = 0)/E_{\Gamma X}(\theta = 1/2)$
ratio < 1: Candidate 2 worse
$\kappa = 0.2$
$\kappa = 0.7$
$\kappa = 2$
3 slowest eigenvectors
Gaussian X
$s_R = \xi r_c$ ($s_F = 6.5$)